

Description of entities:

Entity 1: recipe				
Contains information about recipe which are gathered in our service. An entry is added when a new recipe is added.				
Attributes				
Name	Primary key	Type	Domain	Description
recipe_id	yes	integer	Number from 1 to 10000	Unique recipe identifier.
general_recipe_id	no	Integer	Number from 1 to 10000	Reference to general recipe identifier.
name	no	text	Up to 50 chars. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Name of recipe.
description	no	text	Up to 500 chars. Roman alphabet with polish signs, special characters allowed.	Description of the recipe.
diet_type_id	no	integer	Number from 1 to 100	Reference to diet_type identifier.

Entity 2: general_recipe				
Contains information about recipe. An entry is added when a new recipe is added which general version does not exist in the database.				
Attributes				
Name	Primary key	Type	Domain	Description
general_recipe_id	yes	integer	Number from 1 to 10000	Unique recipe identifier.
name	no	text	Up to 50 chars. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Name of recipe.
photo_url	no	text	Roman alphabet without polish signs, special characters not allowed.	Link to photo of the dish.
meal_type	no	text	One from list: 'BREAKFAST', 'LUNCH', 'DINNER', 'SUPPER', 'DESSERT'	Type of the meal depending on the time of day.
description	no	text	Up to 500 chars.	Description of the recipe.

			Roman alphabet with polish signs, special characters allowed.	
steps	no	text	Up to 500 chars. Roman alphabet with polish signs, special characters allowed.	Instructions on how to prepare a meal.
base_recipe_id	no	integer	Number from 1 to 10000.	Reference to recipe identifier.

Entity 3: diet_type				
Contains information about diet.				
Attributes				
Name	Primary key	Type	Domain	Description
diet_type_id	yes	integer	Number from 1 to 100.	Unique diet identifier
diet_type	no	text	Up to 20 chars. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Name of diet.

Entity 4: contains				
Contains information about recipe's all ingredients, which are individual case for each recipe. An entry is added when the new ingredient is added to recipe.				
Attributes				
Name	Primary key	Type	Domain	Description
contains_id	yes	integer	Number from 1 to 10000.	Unique contains identifier.
Amount	no	float	Number from 0 up to 1000.	Amount of unit needed
Measure	no	text	Number from 0 to 1000, n/a if empty.	Indicates the measurement of the given unit in grams.
recipe_id	no	number	Number from 1 to 10000.	Reference to recipe identifier.
ingredient_id	no	number	Number from 1 to 10000.	Reference to ingredient identifier.

Entity 5: ingredient				
Number of ingredients is up to 10000.Contains information about ingredient used in recipe. An entry is added when a new ingredient is added.				
Attributes				

Name	Primary key	Type	Domain	Description
Id_ingredient	yes	Integer	Number up to 10000.	Unique ingredient identifier.
name	no	text	Up to 50 chars. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Name of ingredient
fat	no	Float	Float from 0 up to 100.	Amount of fat for 100g.
protein	no	Float	Float from 0 up to 100.	Amount of protein for 100g.
carbohydrate	no	Float	Float from 0 up to 100.	Amount of carbohydrates for 100g.
kcal	no	Float	Float from 0 up to 900.	Amount of kcal for 100g.
is_vegan	no	Binary	Contain 0(if not) and 1 if (yes)	Indicates whether it is vegan or not.
is_vegetarian	no	Binary	Contain 0(if not) and 1 if (yes)	Indicates whether it is vegetarian or not.
is_gluten_free	no	Binary	Contain 0(if not) and 1 if (yes)	Indicates whether it is gluten free or not.
is_lactose_free	no	Binary	Contain 0(if not) and 1 if (yes)	Indicates whether it is lactose free or not.
is_keto	no	Binary	Contain 0(if not) and 1 if (yes)	Indicates whether it is keto or not.

Entity 6: substitute				
Contains information about substitutes of given ingredient.				
Attributes				
Name	Primary key	Type	Domain	Description
substitute_id	yes	integer	Number up to 10000.	Unique substitute identifier.
ingredient1_id	no	integer	Number up to 10000.	Reference to ingredient identifier.
ingredient2_id	no	integer	Number up to 10000.	Reference to ingredient identifier.

Entity 7: specific				
Contains information about specific preferences of user.				
Attributes				
Name	Primary key	Type	Domain	Description

specific_id	yes	integer	Number up to 10000.	Unique specific preference identifier
user_id	no	integer	Number up to 1000.	Reference to user identifier.
ingredient_id	no	integer	Number up to 10000.	Reference to ingredient identifier.
likes	No	Binary	Contain 0(if not) and 1 if (yes)	Indicates whether the user likes it or not

Entity 8: review				
Contains information about reviews, which is published by user on a given recipe.				
Attributes				
Name	Primary key	Type	Domain	Description
review_id	yes	integer	Number up to 10000.	Unique specific preference identifier
user_id	no	integer	Number up to 10000.	Reference to user identifier.
general_recipe_id	no	integer	Number up to 10000.	Reference to general_recipe identifier.
rating	No	Integer	Number from 1 to 5	Indicates the degree of liking the recipe by user.
opinion	No	Text	Up to 500 signs. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Describes the experience of recipe.

Entity 9: user				
Contains information about users.				
Attributes				
Name	Primary key	Type/domain	Domain	Description
user_id	Yes	integer	Number from 1 to 10000	Unique user identifier
name	no	text	Up to 20 chars. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Name of user
email	No	Text	Roman alphabet up to 50 chars, must contain @ in between 2 strings.	Email of user
password	No	Text	Minimum 5 and up to 20 chars. Roman alphabet with polish signs, any special	Password of user

			characters are allowed(ex. @,!,_), but not required.	
birthdate	no	date	Allows to introduce date from the interval (1.01.1900-today) The form: ('YYYYMMDD')	Birthdate of the user.
create_date	no	date	The user's account creation date. Allows to introduce date from the interval (1.01.2024-today) The form: ('YYYYMMDD')	Date of the creation of account.
pref_is_vegan	No	Binary	Contain 0(if not) and 1 if (yes)	Specification of being vegan or not.
pref_is_gluten_free	No	Binary	Contain 0(if not) and 1 if (yes)	Specification of eating gluten-free or not.
account_enabled	No	Binary	Contain 0(if not) and 1 if (yes)	Information whether enabled.
account_expired	No	Binary	Contain 0(if not) and 1 if (yes)	Information whether expired.
account_locked	No	Binary	Contain 0(if not) and 1 if (yes)	Information whether locked.
credentials_experied	No	Binary	Contain 0(if not) and 1 if (yes)	Information whether credentials expired.

Entity 10: favourites				
Contains information about user's favourites recipes.				
Attributes				
Name	Primary key	Type	Domain	Description
favourite_id	yes	integer	Number 1 from to 10000.	Unique favourites identifier
user_id	no	integer	Number 1 from to 10000.	Reference to user identifier.
general_recipe_id	no	integer	Number 1 from to 10000.	Reference to general_recipe identifier.

Entity 11: modified_recipe				
Contains the relationship between recipe and user.				
Attributes				
Name	Primary key	Type	Domain	Description
modified_recipe_id	yes	integer	Number 1 from to 10000.	Unique identifier od modified_recipe.
user_id	no	integer	Number 1 from to 10000.	Reference to user identifier.
recipe_id	no	integer	Number 1 from to 10000.	Reference to recipe identifier.

Entity 12: user_diet_type				
Contains the relationship between diet_type_id and user.				
Attributes				
Name	Primary key	Type	Domain	Description
user_id	yes	integer	Number 1 from to 10000.	Reference to user identifier.
diet_type_id	yes	integer	Number 1 from to 10000.	Reference to diet_type identifier.

Entity 13: roles				
Contains the information about the roles and.				
Attributes				
Name	Primary key	Type	Domain	Description

role_id	yes	integer	Number 1 from to 10000.	Unique identifier.
user_id	no	integer	Number 1 from to 10000.	Reference to user identifier.
role	no	text	Up to 20 chars. Roman alphabet with polish signs, any special characters not allowed(ex. @,!,_)	Name of a role.

Relationships

ID	Name	Table 1	Table 2	Type	Description
1	user_diet_type_user	users	user_diet_type	1..n	Defines the diet preferences of each user.
2	user_diet_type_diet_type	diet_type	user_diet_type	1..n	Categorizes users based on specific dietary types.
3	user_roles	users	roles	1..n	Assigns roles to users for system functionalities.
4	general_recipe_recipe	general_recipe	recipe	1..n	Links general recipes to their detailed versions.
5	diet_type_recipe	diet_type	recipe	1..n	Associates recipes with compatible dietary types.
6	recipe_contains	recipe	contains	1..n	Specifies the ingredients required for each recipe.

7	contains_ingredient	contains	ingredient	1..1	Details the nutritional composition of ingredients.
8	substitute_ingredient1	ingredient	substitute	1..n	Allows substitution of ingredients in recipes.
9	substitute_ingredient2	ingredient	substitute	1..n	Defines alternative ingredients for specific recipes.
10	review_user	users	review	1..n	Captures user feedback and ratings for recipes.
11	review_general_recipe	general_recipe	review	1..n	Collects reviews for individual general recipes.
12	favorite_user	users	favorite	1..n	Tracks the favorite recipes of users.
13	favorite_recipe	general_recipe	favorite	1..n	Shows which recipes are marked as favorite by users.
14	modified_user	users	modified_recipe	1..n	Tracks customized recipes created by users.
15	modified_recipe	recipe	modified_recipe	1..n	Maps custom recipes to original recipes.
16	specific_user	users	specific	1..n	Stores user preferences for specific ingredients.
17	specific_ingredient	ingredient	specific	1..n	Indicates user preferences for or against certain ingredients.
18	recipe_general_recipe	recipe	general_recipe	1...1	Links a recipe to its base , general version.

RDB schema

diet_type(diet_type_id, diet_type)

- **Primary Key:** (diet_type_id)

general_recipe(general_recipe_id, name, photo_url, meal_type, description, steps, base_recipe_id)

- **Primary Key:** (general_recipe_id)
- **Foreign Key:** (base_recipe_id) REFERENCES recipe(recipe_id)

users(user_id, user_name, email, password, birthdate, create_date, photo_url, pref_is_vegan, pref_is_gluten_free, account_enabled, account_expired, account_locked, credentials_expired)

- **Primary Key:** (user_id)

ingredient(ingredient_id, name, fat, protein, carbohydrate, kcal, is_vegan, is_vegetarian, is_gluten_free, is_lactose_free, is_keto)

- **Primary Key:** (ingredient_id)

recipe(recipe_id, general_recipe_id, name, description, diet_type_id)

- **Primary Key:** (recipe_id)
- **Foreign Key:** (general_recipe_id) REFERENCES general_recipe(general_recipe_id)
- **Foreign Key:** (diet_type_id) REFERENCES diet_type(diet_type_id)

contains(contains_id, amount, measure, recipe_id, ingredient_id)

- **Primary Key:** (contains_id)
- **Foreign Key:** (recipe_id) REFERENCES recipe(recipe_id)
- **Foreign Key:** (ingredient_id) REFERENCES ingredient(ingredient_id)

substitute(substitute_id, ingredient1_id, ingredient2_id)

- **Primary Key:** (substitute_id)
- **Foreign Key:** (ingredient1_id) REFERENCES ingredient(ingredient_id)
- **Foreign Key:** (ingredient2_id) REFERENCES ingredient(ingredient_id)

specific(specific_id, user_id, ingredient_id, likes)

- **Primary Key:** (specific_id)
- **Foreign Key:** (user_id) REFERENCES users(user_id)
- **Foreign Key:** (ingredient_id) REFERENCES ingredient(ingredient_id)

review(review_id, user_id, general_recipe_id, rating, opinion)

- **Primary Key:** (review_id)
- **Foreign Key:** (user_id) REFERENCES users(user_id)
- **Foreign Key:** (general_recipe_id) REFERENCES general_recipe(general_recipe_id)

favorite(favorite_id, user_id, general_recipe_id)

- **Primary Key:** (favorite_id)
- **Foreign Key:** (user_id) REFERENCES users(user_id)
- **Foreign Key:** (general_recipe_id) REFERENCES general_recipe(general_recipe_id)

roles(role_id, user_id, role)

- **Primary Key:** (role_id)
- **Foreign Key:** (user_id) REFERENCES users(user_id)

user_diet_type(user_id, diet_type_id)

- **Primary Key:** (user_id, diet_type_id)
- **Foreign Key:** (user_id) REFERENCES users(user_id)
- **Foreign Key:** (diet_type_id) REFERENCES diet_type(diet_type_id)

modified_recipe(modified_recipe_id, user_id, recipe_id)

- **Primary Key:** (modified_recipe_id)
- **Foreign Key:** (user_id) REFERENCES users(user_id)
- **Foreign Key:** (recipe_id) REFERENCES recipe(recipe_id)